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DESIGN, IMPLEMENTATION, AND MICROSECOND-LEVEL PERFORMANCE ANALYSIS OF A
JUELS-SUDAN BIOMETRIC FUZZY VAULT OVER $F(2^{31} - 1)$ IN C++17

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Author: Chirantan Ghosh

Affiliation: Department of Computer Science & Information Systems, BITS Pilani

ABSTRACT

Traditional cryptographic hash primitives (such as SHA-256) enforce exact bit-matching and exhibit the avalanche effect, rendering them incompatible with the non-deterministic spatial noise inherent to biometric acquisitions. Storing unencrypted biometric minutiae vectors in databases introduces severe non-revocability hazards. This paper presents a deterministic, high-throughput software implementation of the Juels-Sudan Fuzzy Vault cryptosystem authored in C++17. By evaluating polynomial bindings over the Mersenne prime Galois Field $F(2^{31} - 1)$, leveraging Fermat's Little Theorem for $O(\log P)$ modular inversions, and introducing a 16-bit Cyclic Redundancy Check (CRC-16) integrity gate, the system achieves an empirical False Accept Rate (FAR) of 0.000% while recovering secret keys in 27.2 microseconds on commodity x86-64 hardware.

1. INTRODUCTION & THREAT MODEL

Biometric authentication introduces a fundamental paradox in modern security engineering: unlike passwords, biological signatures cannot be revoked or re-issued once compromised. Naive architectures that store raw minutiae coordinates (x, y) expose users to permanent identity theft upon database exfiltration.

The Juels-Sudan Fuzzy Vault addresses this by binding a cryptographic secret to a noisy

biometric template, camouflaging the genuine features within a dense cloud of random decoy points ("chaff"). The system guarantees zero-leakage of the biometric template in persistent storage.

2. MATHEMATICAL ARCHITECTURE & GALOIS FIELD ARITHMETIC

To prevent compiler-dependent floating-point rounding errors across CPU architectures, all arithmetic operations are strictly bounded to the finite field $F(P)$, where:

$$P = 2^{31} - 1 = 2147483647 \text{ (Mersenne Prime)}$$

A. Polynomial Embedding

A secret master key S is embedded into the baseline coefficient c_0 of a degree- k polynomial:

$$p(x) = c_0 + c_1x + c_2x^2 + \dots + c_kx^k \pmod{P}$$

B. Modular Inversion Optimization

Rather than utilizing branch-heavy Euclidean algorithms, field division is executed via Fermat's Little Theorem:

$$b^{-1} = b^{(P-2)} \pmod{P}$$

Using binary exponentiation, this executes deterministically in 31 bitwise multiplication cycles.

C. Reed-Solomon Reconstruction via Lagrange Interpolation

A query set Q matching at least $(k + 1)$ points on $p(x)$ reconstructs the secret via:

$$c_0 = \text{SUM} [y_i * \text{PROD} (-x_j / (x_i - x_j))] \pmod{P}$$

3. INTEGRITY GATING (CRC-16)

Because any arbitrary subset of $(k + 1)$ points on a 2D plane can mathematically form a valid degree- k polynomial, raw Lagrange interpolation results in false positive decryptions when supplied with random imposter chaff. To eliminate this vulnerability, a 16-bit

Cyclic Redundancy Check (CRC-16-CCITT) checksum of the secret key is encoded alongside `c_0`.

Reconstructed candidates failing the CRC verification are discarded in 4.1 microseconds.

4. EMPIRICAL BENCHMARKS (x86-64, GCC 13.2 -O3)

Metric	Benchmark Value
Vault Enrollment Latency (166 pts)	99.4 μ s
Legitimate Key Recovery ($k = 3$)	27.2 μ s
Imposter Cutoff Latency	4.1 μ s
False Accept Rate (FAR)	0.000% (Over 10^5 imposter iterations)
Vault Memory Footprint	1.32 KB (Contiguous coordinate arrays)
Combinatorial Security Space	$C(166, 4) = 30,820,915$ candidate subsets

5. CONCLUSION

The developed C++17 engine demonstrates that noise-tolerant biometric cryptography can be deployed with zero persistent template leakage while maintaining sub-30 microsecond decoding speeds suitable for high-throughput authentication gateways.